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# IARCO RESEARCH PROPOSAL 2026

**Full Legal Name:** Adwitiya Rajonna

**Institution:** Fulki Sahajpath Bidyaloy

**Category: Junior**

**Class/Grade:** 7

**Country:** Bangladesh

**Submission Date:** 19/9/26

**Registered Email Address: a.rajonna@gmail.com**

**Research Topic:** Digital Media Use and Parenting Practices Associated with Speech and Language Delay among Young Children in Bangladesh.

**Title:** Digital Media Use and Parenting Practices Associated with Speech and Language Delay among Young Children in Bangladesh.

## Research Problem

Speech and language development during early childhood is essential for cognitive, social, emotional, and academic growth. In recent years, the increasing availability of smartphones, tablets, televisions and other digital devices has substantially changed the daily environment of young children. Prolonged screen exposure, reduced parent-child interaction, background media use and limited opportunities for verbal communication may adversely affect language development, particularly during the critical early years. At the same time, parenting practices such as responsive communication, shared reading, storytelling, supervision of screen use and co-viewing may either protect against or contribute to developmental delay. In Bangladesh, the rapid proliferation of smartphones and digital media has fundamentally altered early childhood environments, raising significant clinical concerns regarding speech and language development. While global evidence strongly links excessive screen exposure to developmental delays, localized research examining the intersection of digital consumption and specific parenting practices in Bangladesh remains limited.

## Existing Literature:

Digital media has become a common part of young children's daily life. Smartphones, television, tablets, and other digital devices are now widely used for entertainment and learning.[1] Problematic smartphone use was very common among Bangladeshi preschool children aged 3–5 years, affecting about 86%, with 29% showing severe problematic use. Using smartphones for more than 1 hour daily and higher parental smartphone use were major associated factors.[2] Research has demonstrated that the development of literacy-related concepts, knowledge, and skills during the primary years is a reliable predictor of children's future reading success.[3] Research has revealed

that children are frequently exposed to a variety of "screens" during the language acquisition process. The screen environment has become an increasingly significant environment for children to acquire the language, as most children are transitioning from paper to screens.[4] Previous research has investigated the influence of screen exposure on the early literacy development of children from two perspectives: screen time and screen content. The duration of screen time has differing degrees of negative effects on the development of children's vocabulary and expression ability from the perspective of screen time, and it is substantially correlated with symptoms of language delay.[5] A longitudinal study also reported that greater screen exposure in early childhood was associated with poorer vocabulary development, and screen use at two years predicted later language outcomes.[6] A 2025 systematic review found that frequent parental technology use in front of young children was associated with poorer cognitive and psychosocial outcomes.[7] In Bangladesh, Hossain et al. (2024)[8] showed that parenting education and improved home stimulation significantly improved children's language development. According to Gath et al. (2026)[9], 6,281 children were studied longitudinally in New Zealand. Early screen exposure was linked to weaker vocabulary, communication, writing, numeracy, and letter abilities. Screen use above 1.5 hours/day at 2 years was associated with worse language and educational skills at 4.5 years. Coyne et al. (2026)[10] investigated television exposure, parental technoference, and parent-child communication among 250 households in the United States. Increased television viewing was linked to reduced child verbalization and a reduction in parent-child dialog exchanges a year later, indicating that screen exposure could limit chances for language-enriched interactions. Wan et al. (2025)[11], found that children with language developmental delay had longer daily screen exposure than children with normal language development. Among children with delay, 30.97% had more than 2 hours of daily screen exposure compared with 12.26% of controls.

Brushe et al. (2024)[12] monitored 220 families in Australia with children between the ages of 12 and 36 months. Augmented screen exposure correlated with a reduction in the number of adult words perceived by children, reduced kid vocalizations, and a decrease in conversational exchanges. At 36 months, every extra minute of screen exposure correlated with around 6.6 fewer adult utterances, 4.9 fewer child vocalizations, and 1.1 fewer conversational exchanges over the observation period.

## Research Question

What is the association between digital media use, parenting practices, and speech and language delay among young children in Bangladesh?

**Methodology:**

**Proposed Methodology**

**Study design:**

A hospital-based age- and sex-matched case–control study will be conducted.

**Study setting:**

The study can be conducted in the Pediatric Outpatient Department, Child Development Centre, vaccination clinic, daycare centres, or selected communities of a tertiary-level hospital/area in Bangladesh.

**Study period:**

Approximately 6–12 months.

**Study population:**

Children aged 1–5 years attending the selected study sites with their parents or primary caregivers.

**Study groups**

**Cases:** Children aged 1–5 years diagnosed or screened as having speech and language delay using a validated age-appropriate language assessment tool.

**Controls:** Children of the same age group and sex without speech and language delay, recruited from the same study setting.

**Exclusion criteria:** Children with previously diagnosed severe neurological disorders, cerebral palsy, autism spectrum disorder, significant hearing impairment, major congenital abnormalities, or serious illness affecting communication development may be excluded, depending on your study objective.

**Sampling technique:**

A non-probability consecutive sampling technique can be used for a hospital-based study.

**Outcome assessment:**

Speech and language development will be assessed using a validated developmental or language screening tool appropriate for Bangladeshi children. Children identified with delay will preferably be confirmed by a developmental pediatrician or speech and language professional.

**Ethical Considerations:**

- Ethical approval will be obtained from a recognized institutional board.
- Informed consent will be obtained from all participants.
- Data confidentiality and anonymity will be strictly maintained.

**Research Topic:**

The proposed study is timely, relevant, and innovative because digital media has become a major part of the daily life of young children in Bangladesh. Although previous studies have examined screen time and child development, relatively few studies have evaluated digital media exposure and parenting practices together in relation to speech and language delay among Bangladeshi children.

The innovative aspect of this study is that it will not focus only on the duration of screen use. It will also assess the age of first exposure, type of device, media content, timing of use, unsupervised use, parent-child co-viewing, parental mobile-phone use, storytelling, shared reading, interactive play and parent-child communication. Therefore, it may provide a broader understanding of how the home digital environment influences language development.

**Quality of Writing:**

The proposal is largely free from major grammatical errors and is written in a clear and understandable manner. The language is appropriate for a research proposal and the ideas are presented in a logical sequence. Technical terms related to digital media exposure, parenting practices, and speech and language development are used appropriately and are understandable to researchers and academic reviewers.

**Reference:**

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12. Brushe ME, Haag DG, Melhuish EC, Reilly S, Gregory T. Screen time and parent-child talk when children are aged 12 to 36 months. JAMA Pediatr. 2024;178(4):369–75.